

HAIYANG, China

WMO: 548630

Lat: 36.7750N

Lon: 121.1782E

Elev: 131

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.7	19.6	-1.0	5.2	30.2	2.1	6.1	28.7	23.8	30.7	21.8	31.5	6.2	290	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	88.1	75.6	85.9	75.1	84.0	74.6	80.6	84.2	79.5	82.9	78.3	81.7	8.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.6	154.5	83.0	78.4	148.4	82.0	77.3	142.7	80.8	44.6	84.5	43.3	83.1	42.1	81.6	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.6	18.7	16.1		DB	13.1	92.8	3.0	3.1	11.0	95.0	9.2	96.8	7.5	98.6	5.3	100.8
				WB	11.4	81.8	2.8	1.6	9.4	83.0	7.7	83.9	6.2	84.8	4.2	85.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	30.5	34.1	42.7	53.6	64.1	71.4	77.7	78.4	71.2	60.4	47.0	35.0
	DBStd	17.53	5.71	5.84	6.05	5.40	5.10	3.91	3.51	3.46	4.17	5.72	7.18	6.10
	HDD50	1920	604	445	238	25	0	0	0	0	0	4	140	464
	HDD65	4688	1069	865	691	345	79	3	0	0	5	162	540	929
	CDD50	3971	0	0	11	132	436	641	860	879	637	325	50	0
	CDD65	1263	0	0	0	2	50	194	395	414	191	17	0	0
	CDH74	8498	0	0	0	12	281	799	2937	3531	891	47	0	0
	CDH80	1854	0	0	0	3	53	163	655	864	113	3	0	0
Wind	WSAvg	7.1	7.1	7.2	7.7	8.1	7.6	6.7	6.6	6.6	6.3	6.6	7.1	6.9
Precipitation	PrecAvg	30.6	0.4	0.3	0.7	1.8	2.4	3.3	8.1	7.7	3.3	1.4	0.9	0.4
	PrecMax	65.4	2.0	1.4	2.8	5.0	7.8	9.8	19.6	24.1	13.5	4.1	5.5	1.3
	PrecMin	15.4	0.0	0.0	0.0	0.0	0.2	0.2	1.9	1.3	0.1	0.0	0.0	0.0
	PrecStd	9.8	0.5	0.4	0.7	1.3	2.0	2.6	4.6	4.9	2.8	1.2	1.1	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	53.0	65.1	73.8	85.5	88.8	91.5	91.1	85.5	78.3	67.3	54.5
		MCWB	41.6	42.4	49.6	56.3	63.0	68.2	77.4	77.4	71.0	63.1	57.3	49.4
	2%	DB	44.6	48.8	59.4	69.1	80.5	84.4	88.2	88.0	82.9	74.6	63.8	50.5
		MCWB	39.0	40.5	47.6	54.7	61.4	67.6	77.6	76.6	69.6	61.6	55.1	44.5
	5%	DB	41.9	45.9	55.3	65.6	76.8	81.2	85.6	86.0	80.8	71.9	61.1	47.5
		MCWB	36.3	39.1	44.6	52.3	60.6	67.2	76.9	76.5	69.0	61.2	54.2	41.9
	10%	DB	39.5	43.4	51.9	62.5	73.6	78.5	83.4	84.3	78.9	69.5	58.3	44.5
		MCWB	34.4	37.7	43.0	51.2	59.6	67.3	76.2	76.5	68.2	60.0	51.7	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	46.7	53.2	61.6	68.4	75.5	82.2	81.9	77.7	68.5	61.6	51.5
		MCDB	45.7	49.2	60.1	69.1	77.8	80.8	87.2	85.8	81.0	73.0	64.6	53.6
	2%	WB	40.1	43.1	49.5	58.2	66.1	73.3	80.5	80.9	74.9	66.1	58.5	46.1
		MCDB	43.1	46.2	55.4	64.2	73.9	76.9	84.5	84.4	78.7	70.9	61.4	49.5
	5%	WB	37.5	40.7	47.3	56.1	64.2	71.8	79.3	80.0	72.9	64.2	55.4	42.8
		MCDB	40.9	44.5	52.4	61.1	70.7	75.8	82.9	83.3	77.4	69.2	59.5	46.6
	10%	WB	34.9	38.5	45.1	54.2	62.6	70.4	78.1	79.0	71.2	62.3	52.4	39.7
		MCDB	38.9	42.4	50.1	59.1	68.6	74.6	81.5	82.3	76.2	67.4	57.7	43.9
Mean Daily Temperature Range	5% DB	MDBR	13.0	13.7	14.4	14.2	14.3	11.3	9.0	10.2	13.3	14.8	14.1	12.8
		MCDBR	15.8	16.4	18.6	19.1	20.5	17.3	12.6	12.1	15.2	16.5	16.0	16.1
		MCWBR	11.1	10.6	10.9	9.8	8.7	7.2	5.8	5.4	6.8	8.5	10.4	11.5
	5% WB	MCDBR	14.5	13.8	16.3	14.8	15.4	11.1	9.1	8.2	11.1	13.4	13.6	14.6
		MCWBR	11.7	10.6	10.8	9.1	8.5	5.9	5.1	4.6	6.6	8.8	11.1	12.0
Clear-Sky Solar Irradiance	taub	0.482	0.538	0.591	0.612	0.636	0.730	0.709	0.634	0.557	0.580	0.549	0.490	
	taud	1.817	1.732	1.655	1.664	1.657	1.532	1.631	1.769	1.878	1.733	1.747	1.821	
	Ebn at Noon	206	212	216	222	220	199	202	214	223	197	183	192	
	Edh at Noon	51	63	74	77	79	89	81	69	58	62	55	48	
All-Sky Solar Radiation	RadAvg	759	1005	1426	1660	1800	1669	1476	1482	1369	1105	797	665	
	RadStd	66	97	103	89	130	135	157	160	109	96	80	45	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+1.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+92	

Nomenclature: See separate page